12 Clustering

This chapter covers

- Classification of different types of clustering
- Partitioning Clustering
- Understanding and implementing k-means
- Density-based clustering
- Understanding and implementing DBSCAN
- OPTICS: refining DBSCAN as a hierarchical clustering
- Evaluating clustering results

In the previous chapters we have described, implemented and applied three data structures designed to efficiently solve nearest neighbor search; when we moved to their applications, we mentioned that clustering was one of the main areas where an efficient nearest search could make a difference. So far we had to delay this discussion, but now it’s finally time to put the icing on the cake, and get the best of our hard work. In this chapter, we will first briefly